Shreyash Tripathy

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Education

Kalinga Institute of Industrial Technology, Bhubaneswar — **2024 – Present**
B.Tech in Computer Science & Engineering — *9.70 / 10 CGPA*

DAV Public School, Chandrasekharpur, Bhubaneswar — **2024**
Class XII — *95%*

DAV Public School, Chandrasekharpur, Bhubaneswar — **2022**
Class X — *97%*

Professional Experience

PruTech Solutions, India – *AI Project Intern* — **May 2026 – Aug 2026**

- Engineered **CHITRADARSHI**, an offline multimodal surveillance intelligence system, by evaluating competing vision-language architectures and video inference strategies to establish a GPU-native inference pipeline on **NVIDIA DGX Spark**, meeting enterprise latency, GPU memory, and data-sovereignty requirements.
- Designed a **5-stage, 3-model** multimodal AI pipeline integrating visual reasoning, speech transcription, and incident reasoning to classify **37 anomaly categories**, achieving **55–70 s** end-to-end inference while maintaining **zero false-positive** activations across the evaluated surveillance dataset.

Projects

Multi-Disease Clinical Decision Support Platform — Source

- Architected **4 disease-specific prediction pipelines**, selecting independent ML models through **ROC-AUC–driven** evaluation to deliver patient-specific risk assessment with **sub-500 ms** inference.
- Translated model explanations into structured clinical evidence for LLaMA 3, generating patient-specific recommendations in **2–3 s**, supporting retrieval-backed health guidance, and producing comprehensive clinical reports in **under 2 s**.

Hybrid Career Intelligence and College Recommendation Platform — Source

- Formulated a hybrid recommendation architecture combining a **5-factor weighted** career-matching algorithm, semantic retrieval, and LLM reasoning to generate personalized career guidance, **25**+ learning roadmaps, and JEE-based Dream, Target, and Safe college recommendations from structured and semantic user profiles.
- Validated structured AI outputs through retrieval grounding, prompt constraints, and backend JSON verification, integrating **3 AI pipelines** (career guidance, college prediction, and speech-to-text) into a unified decision workflow supporting **190**+ engineering colleges.

Awards & Achievements

- **Champion, NSS SCE Ideaforge 2K24:** Secured **1st place** among **300**+ teams with **BoldFlushX**, an IoT-enabled smart sanitation system for waste segregation and automation.
- **Qualified for the Nodal Round, Smart India Hackathon 2025:** Ranked **46th** among **750** teams in the university selection process with **Nivaaz**+, a real-time telemedicine consultation platform.
- **Finalist, B-Plan Competition, Kshitij 2025, IIT Kharagpur:** Advanced to the **Top 10** among **300**+ teams, presenting **BoldFlushX** to mentors and venture capitalists.
- **Finalist, Tech Zephyr 2025, IIT Bhubaneswar:** Shortlisted in the **Top 10** among **250** teams for **QuitKit**, a gamified smoking cessation platform.
- **Special Mention (3rd Place), KIIT MUN 2025:** Represented Turkey in UNCSW, securing **3rd place** among **150** delegates and receiving a cash prize of **INR 8,000**.

Technical Skills

- **Programming Languages:** Python, C/C++, Java, JavaScript, TypeScript
- **AI Engineering:** LangChain, Hugging Face, Ollama, ChromaDB, OpenCV, Scikit-learn, XGBoost, LightGBM, SHAP
- **Frameworks & Technologies:** FastAPI, Streamlit, REST APIs, PostgreSQL, Next.js
- **Development Tools:** Git, GitHub, Vercel, Arduino IDE, ESP-IDF

Technical Leadership

KSAC and Outreach Lead and IoT Developer, IoT Labs — **Jan 2025 – Present**

- Integrated sensing, computer vision, and embedded components across **3 IoT systems**, contributing to campus automation through an **ID card-based facial recognition attendance solution**.

AI/ML Member, GeeksforGeeks KIIT Chapter — **Feb 2025 – Present**

- Collaborated on the development of **4 AI applications**, contributing to implementation, testing, and technical discussions across healthcare, surveillance, recommendation, and resume analysis.

Senior Marketing Executive, FED KIIT — **Nov 2024 – Dec 2025**

- Coordinated sponsorship engagement for **6 university events**, partnering with **REMAX Temple City** to support event execution and stakeholder communication.